

UESTC3010: Team Design Project and Skills

General Rules for the Competition Day

1. The team car demonstration and task completion session will take place on **Saturday, 6 June 2026**, and will begin promptly at **9:00 am**.
 - a. All teams presenting in the morning session should report to the venue at **8:30 am**, and all teams presenting in the afternoon session should report to the venue at **13:30**.
 - b. All teams should be prepared to demonstrate their car on time.
 - c. Teams should have the batteries fully charged well before their dedicated time slot. Use of power packs to supplement or replace the battery will not be allowed during the demonstration.
2. Each team will have **one** opportunity to complete all the three tasks during their dedicated time slot. No additional opportunities to complete the tasks will be given later in the day.
 - a. Each team will be allotted **10 minutes** to complete the three tasks. No time will be allocated for debugging; therefore, each team should be ready to start their car as soon as they called to the track. The software should be pre-downloaded onto the car's microcontroller before the allocated timeslot. Any team that fails to start their car within the first **five minutes** of their allocated timeslot will be disqualified.
 - b. It is the team's responsibility to check their scheduled demonstration time and be ready at the starting time. A team that fails to arrive at the assigned track within the 5-minute window will be **dismissed** from the demonstration.
3. The car must operate using a program that has been pre-downloaded onto a microcontroller. Instructions must not be transmitted to the car in real time, and the use of any external computational devices is prohibited. **Any team found to be violating these rules will be dismissed from the demonstration.**
4. There are three tasks to be completed on the track. Task 1 carries a maximum of 50 marks, task 2 carries a maximum of 20 marks and task 3 carries a maximum of 30 marks.
 - a. You will receive full marks for each of the tasks only if you complete it in the first run. You will be penalized by **two marks** for every external interference, touch or restart.
 - b. To receive full marks for the subsequent tasks, the transition from one task to another should be programmed (without any external interference). You will be penalized for any repositioning or restarting (as per the above rule) for the subsequent tasks if it does not transit automatically from one task to another.
5. Electrical systems and all connections to circuit components and subsystems must be rugged and reliable.
 - a. Wires should be soldered onto PCBs, V-board, or punchboard or screwed into terminal blocks on PCBs, V-board or punchboard. Breadboard circuits are not allowed.
 - b. A fuse and ON-OFF switch should be placed between the battery and the rest of the robot. The fuse should be sized appropriately so that it will not be damaged during the normal operation of the car but will create an open circuit should more current than expected be drawn from the battery.
6. Each team is expected to **design a motor driver circuit and the PCB** on which this circuit is constructed.

We stress again that the code must be pre-downloaded onto the micro-controller, controlling your car remotely is strictly prohibited/not allowed. **Any team that controls their car remotely will be dismissed.**